

THOMAS ZHANG

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EDUCATION

Princeton University '28 | **GPA:** 3.96 | Prospective B.S.E. in Electrical and Computer Engineering | Shapiro Prize for Academic Excellence (2025)

Relevant University Coursework: “Computer Architecture”, “Operating Systems”, “Contemporary Logic Design”, “Electronic Circuit Design, Analysis and Implementation”, “Information and Signals”, “Introduction to Programming Systems”, “Algorithms and Data Structures”, “Data Intelligence: Modern Data Science Methods”, “Fundamentals of Statistics”, “Number Theory”, “Single Variable Analysis”, “Multivariable Calculus”

SKILLS:

Digital Design: Verilog, SystemVerilog, RTL Design, Verilog Testbenches, Waveform Debugging, CPU/GPU Microarchitecture, PCI-Express, RISC-V, SoC interconnect; **Software / Embedded:** C/C++, Arduino, FreeRTOS, ESP32, KiCAD, Python, Java, JavaScript/React.js, PostgreSQL, Git, Docker, PCB Layout, Schematic Capture, Oscilloscope & Multimeter Debugging

EXPERIENCE

Researcher - Safety-Oriented Microarchitecture | Princeton Parallel Group **May 2026 - Present**

- Designed a small systolic tensor core on the Vortex RISC-V GPGPU (SystemVerilog), tapping L2 hidden-state traffic to run an on-chip LLM misalignment safety probe. Wrote tiled GEMM kernels in C++ for the Vortex runtime, and built a Python-driven regression suite for verification.
- Implemented a hardware hashing unit (SystemVerilog) to identify AI model weights as they are loaded into GPU VRAM by snooping memory-mapped AXI interconnect bursts.

Research Assistant - Microfabrication | Fu Lab @ Princeton University **August 2025 - Dec 2025**

- Conducted microfabrication procedures (KLayout mask design, photolithography, spin-coating, metal deposition) to build a prototype centimeter-scale optoelectronic biosensing chip for pharmacology and neuroscience applications.

Full-Stack Software Developer, Data Scientist | High Meadows Environmental Institute **Jun 2025 – Aug 2025**

- Built a React + FastAPI + PostgreSQL web app (indiaairpollution.mauzerall.scholar.princeton.edu) visualizing city-level air pollution data in India to quantify the efficacy of government policies over time.
- Dockerized and deployed via a Jenkins pipeline.
- Reduced missing-data bias by parallelizing Monte Carlo simulations on Princeton’s Della cluster.
- Isolated emissions trends from historical weather patterns w/ machine learning (Random Forest).

PROJECTS

Team Lead | Pacbot at Princeton University Robotics Club **Sep 2024 - Present**

- Designed power topology, motor-drive circuitry, IR transceiver analog front-end, and KiCad PCB layout for a differential-drive Pac-Man maze robot; achieved ~4x PID update frequency and ~2x robot velocity over omniwheel design.
- Debugged PCB hardware using oscilloscopes and multimeters; diagnosed and resolved signal integrity and analog circuit issues across multiple board revisions.
- Implemented low-level drivers for TCP/IP communication, encoders, IMU odometry, and PID motor control in ESP-IDF (ESP32/FreeRTOS) for real-time closed-loop navigation.
- Placed 2nd in the 2025 National Pacbot Competition.

Academic Project | Out-of-Order Dual-Issue Processor **Feb 2026 - April 2026**

- Designed and implemented a fully bypassed, pipelined dual-issue RISC-V processor in Verilog, including datapath, control logic, and a pipelined multiplication/division unit.
- Implemented a scoreboard and reorder buffer to support out-of-order execution and dependency tracking. Developed Verilog testbenches and assembly-level verification programs to validate processor correctness.
- Implemented control logic for exclusive and inclusive cache hierarchies.

Academic Project | 16-bit RTL Processor on FPGA **Dec 2025**

- Designed and implemented a 16-bit stored-program processor with a 20-instruction ISA in Verilog. Synthesized and deployed the processor on a Xilinx FPGA using Vivado.